

SUPPLY CHAIN & LOGISTICS ANALYSIS

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1. Project Overview :

The *Supply Chain & Logistics Analysis* project focuses on evaluating and optimizing the efficiency of supply chain operations. It involves analyzing key components such as

1. Procurement & Suppliers.
2. Inventory & Warehouse.
3. Transportation.
4. Order Fulfillment.
5. Performance.

Project Objective:

The primary objective of this project is to enhance the performance and efficiency of the supply chain:

- Analyzing key metrics such as delivery time, transportation cost, inventory levels, and order fulfillment rates.
- Identifying inefficiencies and delays in the supply chain process.
- Improving demand forecasting and inventory optimization.
- Reducing logistics and operational costs.
- Enhancing customer satisfaction through timely and accurate deliveries.
- Supporting data-driven decision-making with dashboards and visual insights.

2. Data Sources:

- **Source Description and Timeline:**
<https://excelx.com/practice-data/generators/supply-chain-logistics/>
Time Line: 2022-2026
- **Domain:** Logistics Retail Company

3. Problem Statement:

- A regional retail company is facing late deliveries, rising inventory costs, and inconsistent supplier lead times across its distribution network.
- Total number of orders delivered.
- Average delivery time.
- Monthly transportation cost.

- Why delivery delays increased in a specific month.
- Forecasting future demand.
- Recommending suppliers for better efficiency.

4. Attribute (Column /Features) Details:

Procurement & Suppliers:

Table Name	Columns
Purchase Orders	PO_ID,SupplierID,OrderDate, ExpectedDeliveryDate,Status, TotalCost
PurchaseOrder Lines	POLineID,PO_ID, MaterialID,Quantity, UnitCost
Supplier Master	SupplierID,SupplierName, Country
RawMaterials Master	MaterialID,MaterialName, Category, UnitCost

Inventory & Warehouse

Table Name	Columns
Warehouse Inventory	InventoryID,MaterialID, WarehouseID,QuantityOnHand, LastStocktakeDate

Stock Movements

MovementID,MaterialID,
WarehouseID,MovementType,
Quantity, Date

Warehouse Master

WarehouseID,WarehouseName,
Location

Transportation:

Table Name

Columns

Shipment Tracking

ShipmentID,PO_ID,
CarrierID,Origin,
Destination,ShipDate,
ActualDeliveryDate,
Status

Carriers Master

CarrierID, CarrierName

Order Fulfillment:

Table Name

Columns

Customer Sales Orders

SalesOrderID,
OrderDate,
ProductID,
ProductName,
Quantity,
CustomerID

Fulfillment Status	FulfillmentID, SalesOrderID, ShipmentID, Status
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Performance:

Table Name	Columns
Supplier Performance	SupplierID,SupplierName, OnTimeDeliveryRate,QualityScore, AverageLeadTime_Days

5. Tools & Technologies

- **Excel:** Data cleaning, Transformation,Power Query.
- **Power BI:** Data modelling, DAX calculations, visualization, and interactive dashboard creation.

6. Data Pre-Processing (Excel / Power Query)

Tasks Performed:

- **Data Cleaning & Transformation:** Removed duplicates, handled missing values, standardized formats, and created calculated fields.
- **Filtering & Sorting:** Organized data to focus on relevant records.
- **Pivot Tables:** Generated Pivot Tables for data summarisation and initial insights.
- Convert the data into Fact and Dimension Table

7. Data Modelling and DAX (Power BI)

- **Data Model:**

☐ From: table (column) ↑	Relationship	To: table (column)	Status	
☐ Customer_Sales_Orders (Sales...	1 — 1	Fulfillment_Status (SalesOrderl...	Active	...
☐ DataTable (Year)	* — *	Purchase_Orders (Year)	Active	...
☐ Fulfillment_Status (ShipmentID)	1 — 1	Shipment_Tracking (Shipmentl...	Active	...
☐ Purchase_Order_Lines (Materia...	* — 1	Raw_Materials_Master (Materi...	Active	...
☐ Purchase_Order_Lines (PO_ID)	* — 1	Purchase_Orders (PO_ID)	Active	...
☐ Purchase_Orders (SupplierID)	* — 1	Supplier_Performance (Supplie...	Active	...
☐ Shipment_Tracking (CarrierID)	* — 1	Carriers_Master (CarrierID)	Active	...
☐ Stock_Movements (MaterialID)	* — 1	Raw_Materials_Master (Materi...	Active	...
☐ Stock_Movements (Warehouse...	* — 1	Warehouse_Master (Warehous...	Active	...
☐ Supplier_Master (SupplierID)	1 — 1	Supplier_Performance (Supplie...	Active	...
☐ Warehouse_Inventory (Materia...	* — 1	Raw_Materials_Master (Materi...	Active	...
☐ Warehouse_Inventory (Wareho...	↓ ↑ ↺	Warehouse_Master (Warehous...	Active	...

Close

Measures:

1) On-Time %

Syntax = **DIVIDE([On-Time Orders],[Total Order Lines of Purchase])**

2) On-Time Orders

Syntax =

CALCULATE(

COUNT(Purchase_Orders[OrderDate]),

Purchase_Orders[OrderDate] <= Purchase_Orders[ExpectedDeliveryDate]

)

3) Delayed Orders

Syntax =

CALCULATE(

COUNTROWS(Purchase_Orders),

Purchase_Orders[OrderDate] <=Purchase_Orders[ExpectedDeliveryDate]

)

4)Cancelled Orders

Syntax =

CALCULATE(COUNTROWS(Purchase_Orders),Purchase_Orders[Delivery Status] = "Cancelled")

5)Final Delivery Status

Syntax =

SWITCH(
TRUE(),
Shipment_Tracking[Status] <> "Delivered", "Pending",
Shipment_Tracking[Delay Days] > 0, "Late",
Shipment_Tracking[Delay Days] = 0, "On Time",
Shipment_Tracking[Delay Days] < 0, "Early"
)

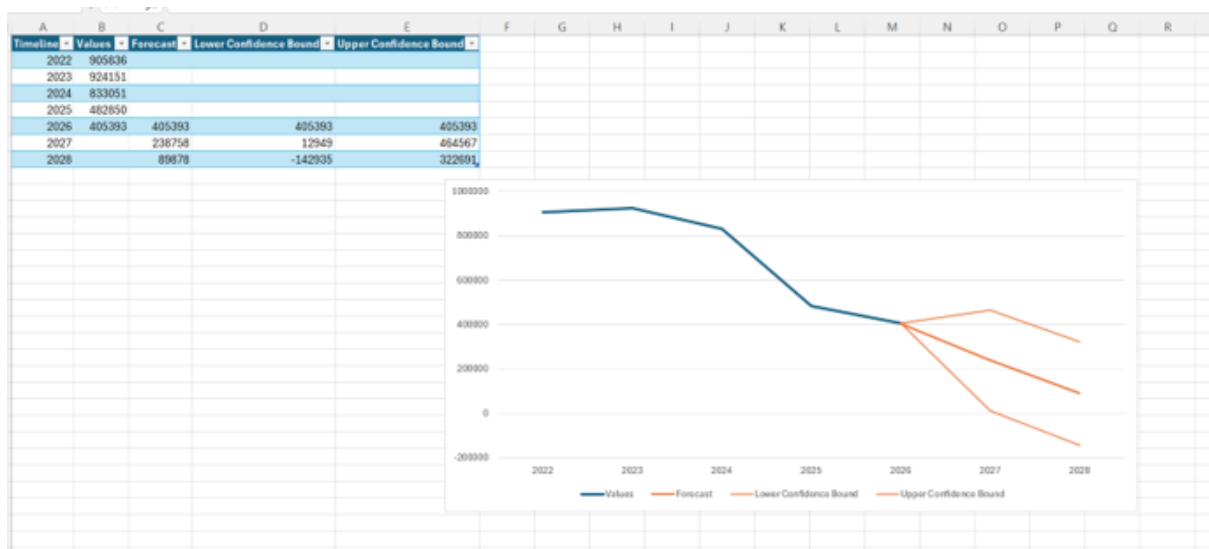
6) Total Shipments

Syntax **=COUNT(Shipment_Tracking[ShipDate].[Month])**

8. Analysis and Visualizations (Power BI)

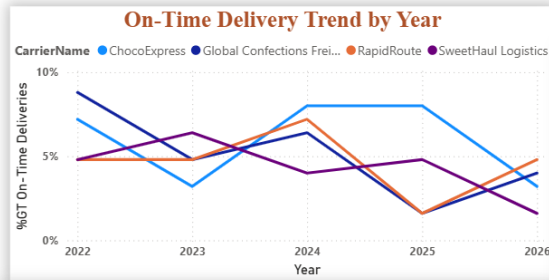
Dashboard Features:

Forecast Future Year :





1. What is the overall on-time delivery rate?



2K

Total Shipments

125

Delayed Shipments

125

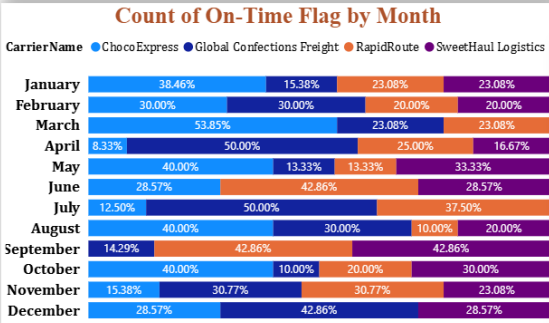
On-Time Deliveries

Delivered Count by CarrierID

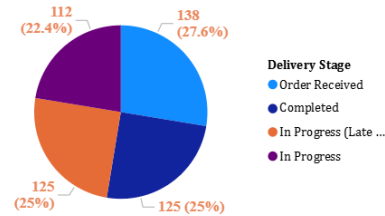


CarrierID

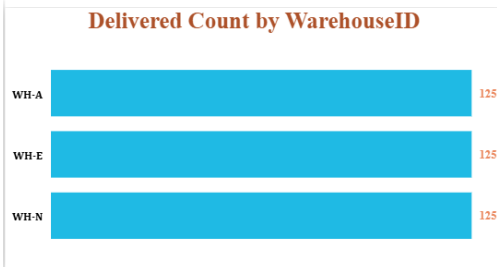
- ☐ CAR-01
- ☐ CAR-02
- ☐ CAR-03
- ☐ CAR-04



Count of ShipmentID by Delivery Stage

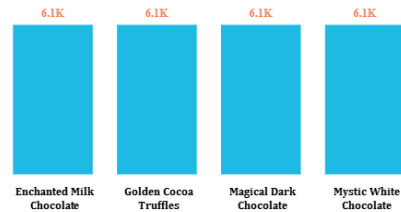


2. Which warehouse has the highest number of delayed shipments?

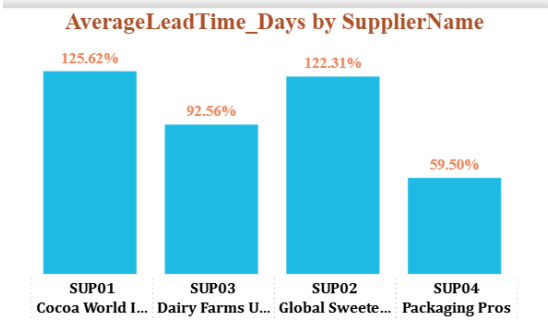


3. What is the average inventory level by product category?

Average Inventory by Category by ProductName



4. Which supplier has the longest average lead time?



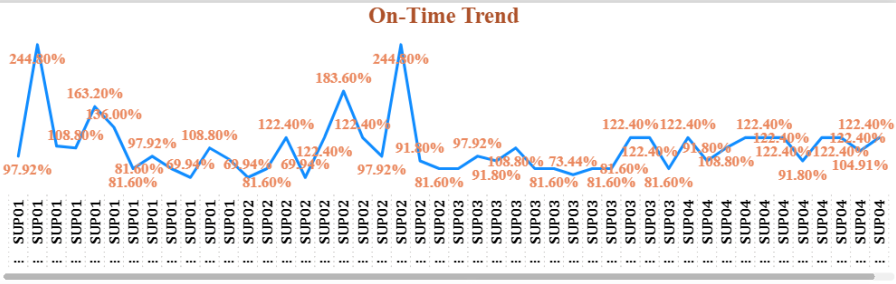
Warehouse Name

- ☐ Asia-Pacific Hub
- ☐ Euro Hub
- ☐ North America Hub

6.10K

Average Inventory by Category

1. Why did on-time delivery rates fall in the most recent month?



1.36K

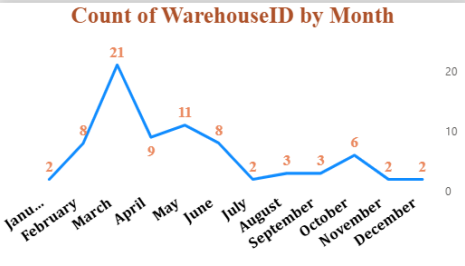
Avg Inbound

77

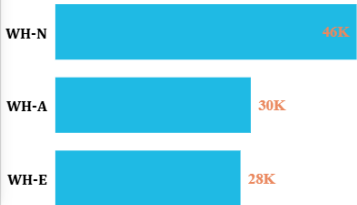
Inbound Count

104K

Total Inbound



Total Inbound by WarehouseID



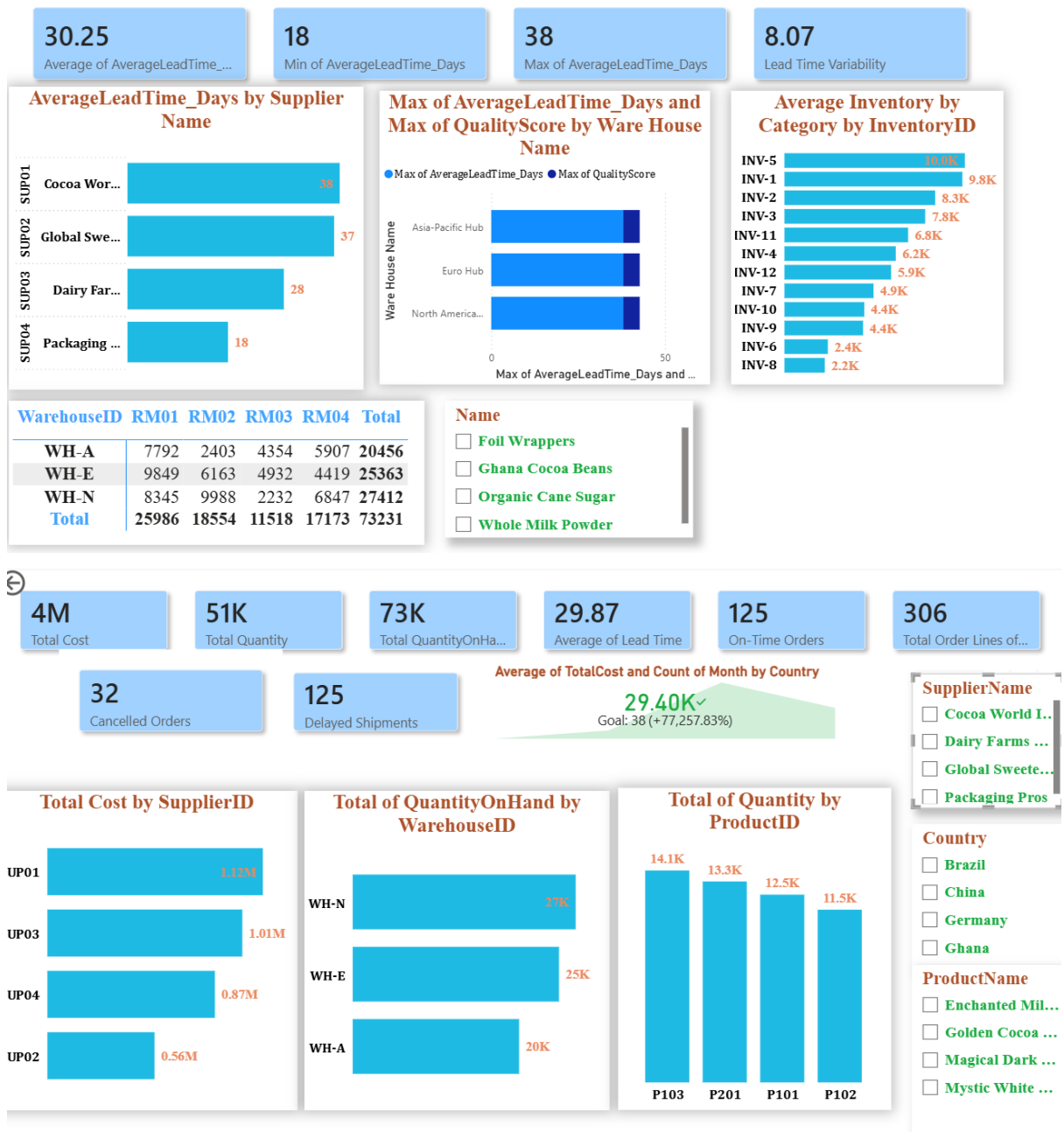
Warehouse Name

- ☐ Asia-Pacific Hub
- ☐ Euro Hub
- ☐ North America Hub

Supplier Name

- ☐ Cocoa World Inc.
- ☐ Dairy Farms United
- ☐ Global Sweeteners
- ☐ Packaging Pros

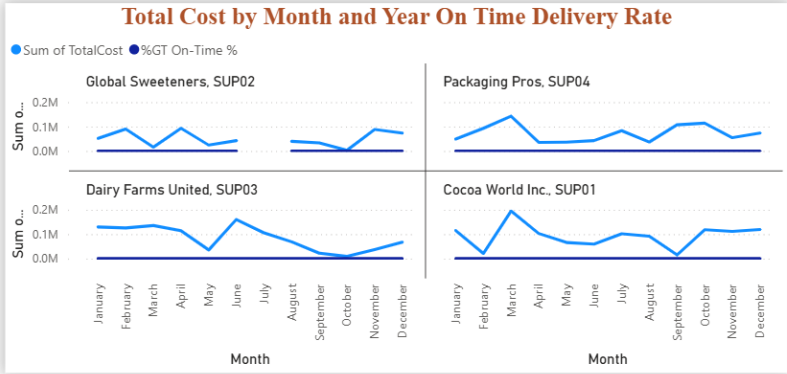
2.What factors are causing high inventory holding costs?

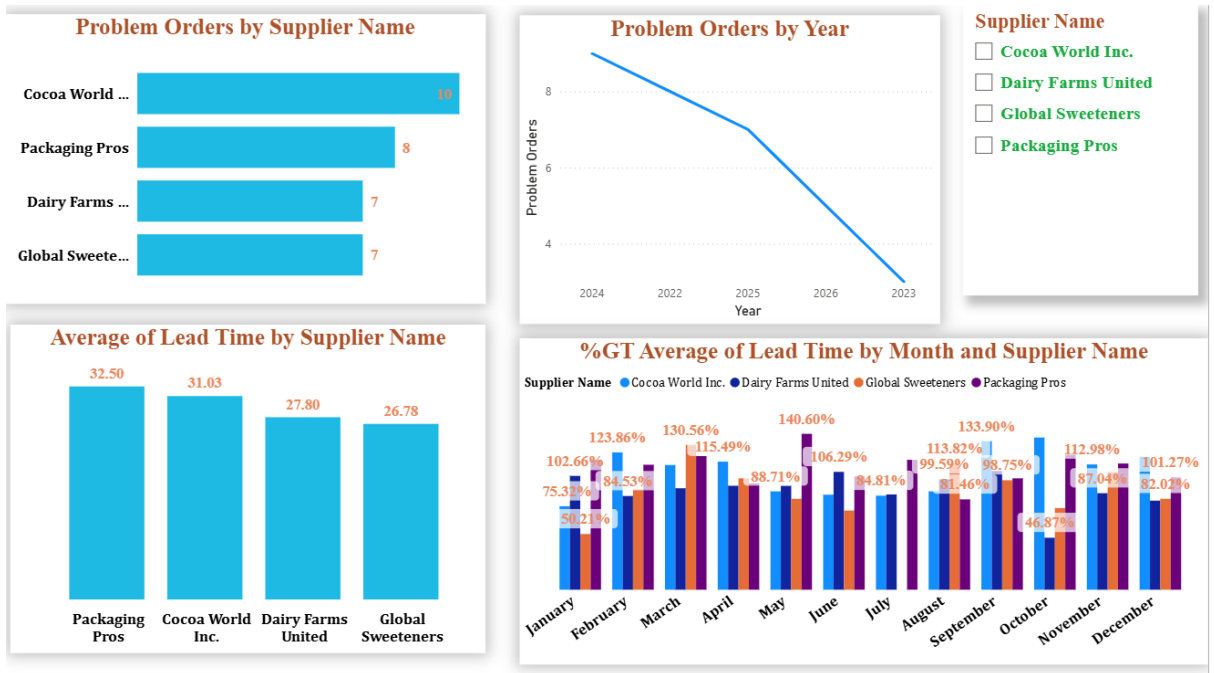


On Time Delivery Rate by Supplier Name						
SupplierName	2022	2023	2024	2025	2026	Total
Cocoa World Inc.	97.92%	91.80%	106.43%	114.24%	146.88%	104.52%
Dairy Farms United	87.43%	78.34%	94.15%	97.92%	89.02%	88.48%
Global Sweeteners	90.19%	122.40%	66.76%	136.00%	97.92%	98.78%
Packaging Pros	81.60%	129.60%	107.10%	146.88%	97.92%	108.09%
Total	90.04%	100.15%	97.14%	117.87%	102.00%	100.00%

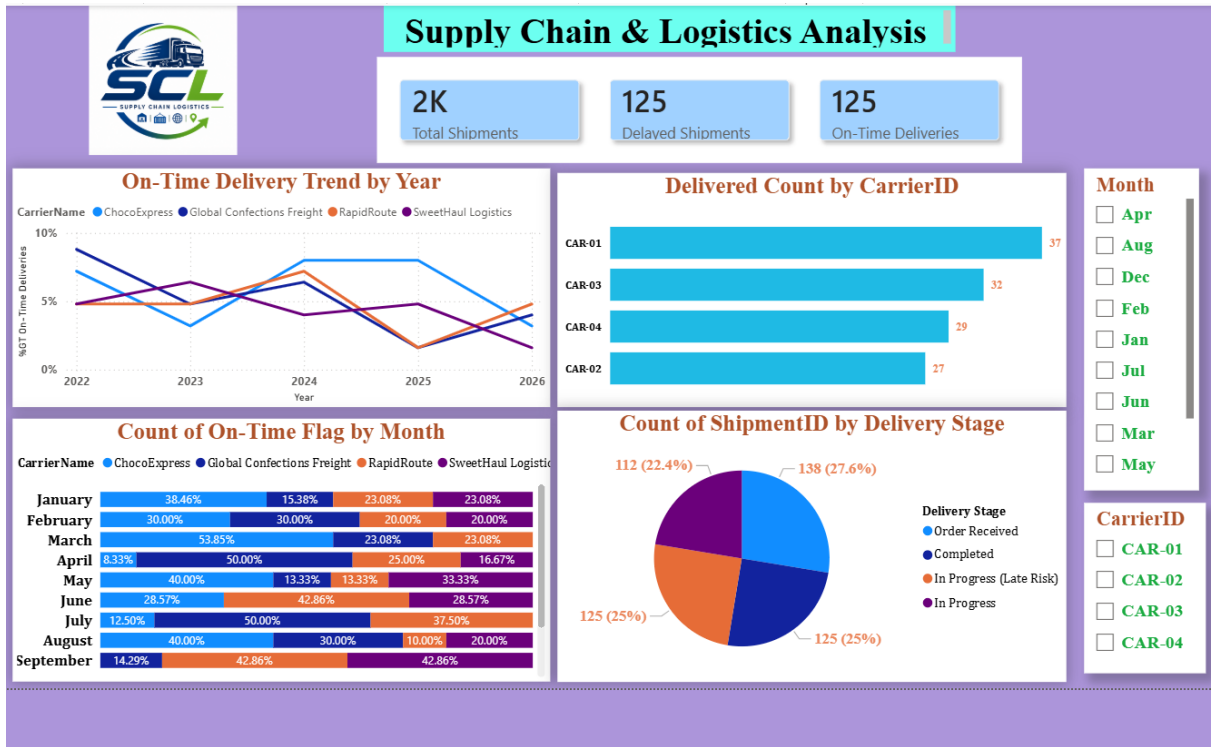
Country

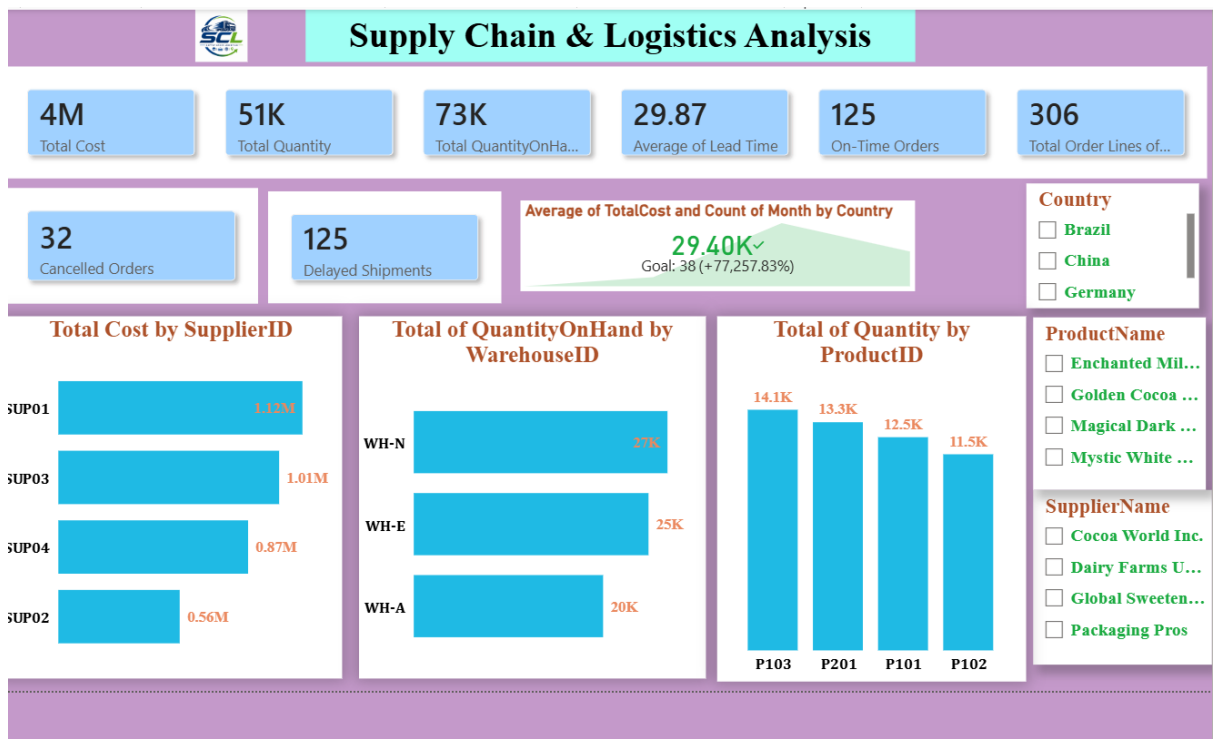
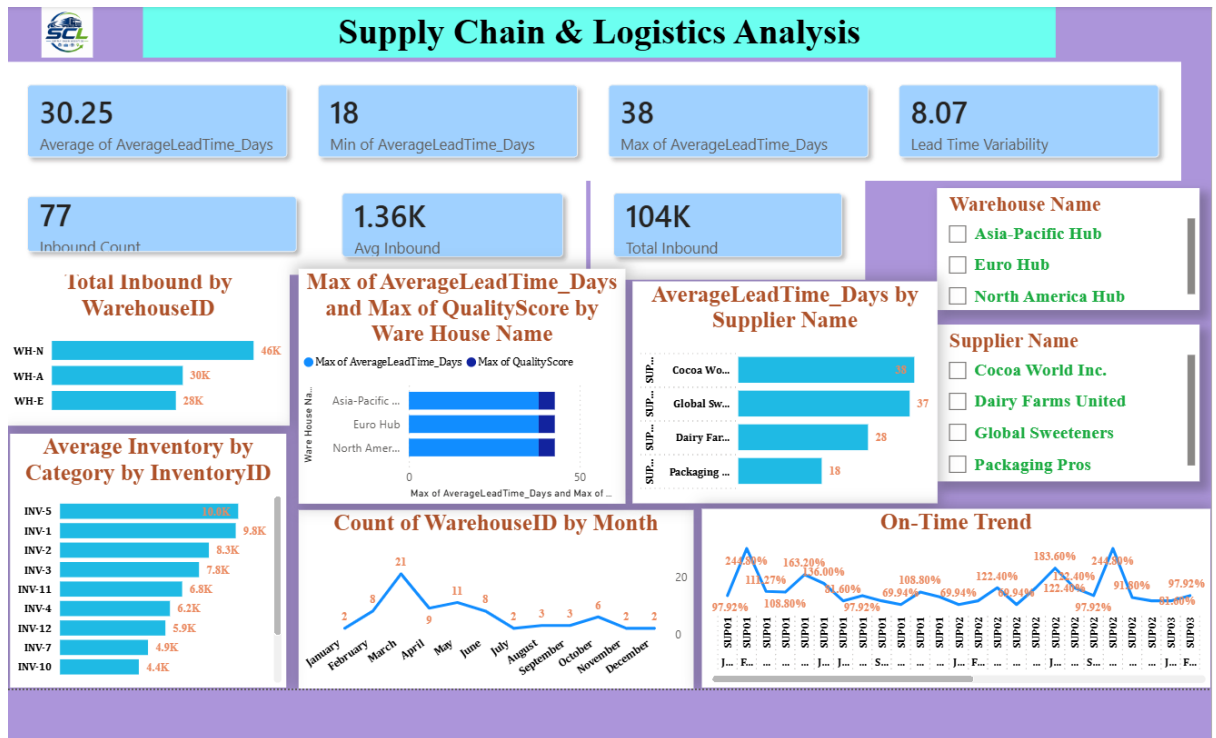
- ☐ Brazil
- ☐ China
- ☐ Germany
- ☐ Ghana





REPORT





9.Final Insights:

- On-time delivery is below target; Warehouse WH-E has the most delays; slow movers hold excess stock; Supplier S4 has longest, most variable lead times.
- Drop in on-time rate caused by supplier lead-time spikes and a local demand surge; WH-E delays due to inbound congestion and understaffing; high holding costs from over-ordering slow SKU(Stock Key Unit).

- Expect continued longer lead times for S4 next month and a small rise in delays; 5–10% stockout risk for top fast SKUs during peak week; excess stock will normalize in 6–8 weeks if no new overorders.
- Immediately surge staff at WH-E and reroute some shipments; renegotiate S4 SLAs and stagger orders; implement per-SKU reorder rules and monitor weekly OTD, lead-time SD, WH-E inbound time, and days of supply.

10. Conclusions

Address immediate WH-E capacity and staffing issues and stabilize S4 lead times through SLA renegotiation and staggered ordering. Implement per-SKU reorder rules and weekly monitoring to reduce holding costs, protect fast SKUs from stockouts, and return OTD to target within the coming weeks.